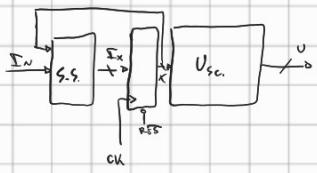
Progetto di MSF

0) Specifiche

1) Descrizione formale in linguaggio ASM — esistono anche (VHDL, Verilog) — Linguaggio di descrizione dell'hw

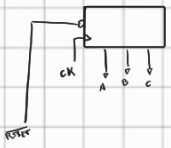
2) Registri di stato

3) Rete combinatoria, stato successivo e Uscite



Ogni x cambia

→ Specifiche

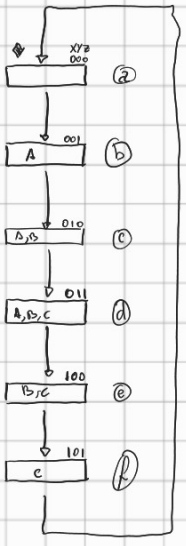


A,B,C = { 1 luce accesa / 0 " spenta



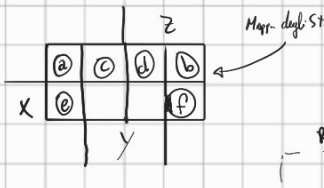
→ Procedimento tutto sistematico

1) Descrizione ASM

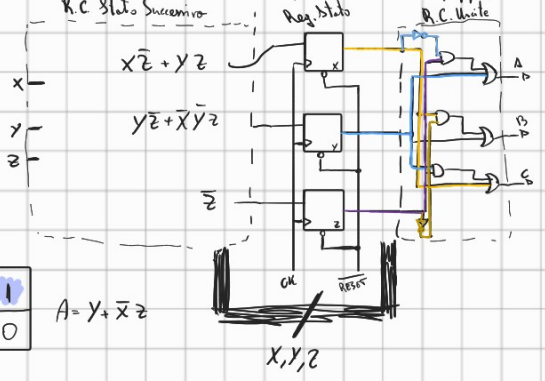


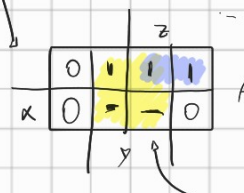
3) Reti Combinatorie
- 3a) Uscite
- 3b) Stato successivo

2) Registro di Stato
- 2a) numero di F.F.
- 2b) Assegnazione degli stati



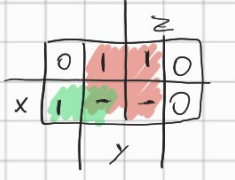
Per ora supponiamo che sono 3 flip flop

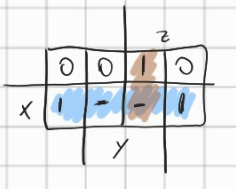




se costruisci con le indifferenze c'è un MAGGIOR pericolo di ERRORI (scelta)

→ non si possono calcolare le USCITE non previste

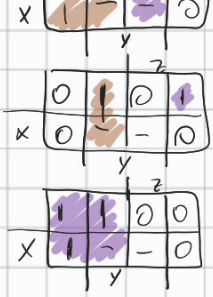




Stato Attuale			Stato Successivo		
a	0	0	0	0	1 b



	x	y	z	x	y	z	
(b)	0	0	1	0	1	0	(c)
(c)	0	1	0	0	1	1	(d)
(d)	0	1	1	1	0	0	(e)
(e)	1	0	0	1	0	1	(f)
(f)	1	0	1	0	0	0	(a)



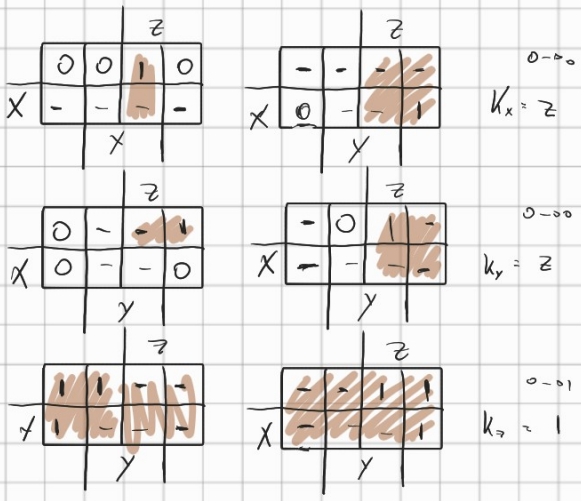
$D_x = y\bar{z} + \bar{x}\bar{y}z$

$D_z = \bar{z}$

FF JK??

Tabella di eccitazione

Transizione	J	K
0→0	0	-
0→1	1	-
1→0	-	1
1→1	-	0



$K_x = z$

$K_y = z$

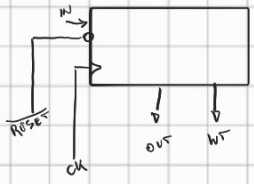
$K_z = 1$

$J_x = yz$ $\quad K_x = z$

$J_y = \bar{x}z$ $\quad K_y = z$

$J_z = 1$ $\quad K_z = 1$

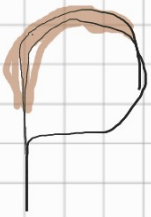
Es. 9

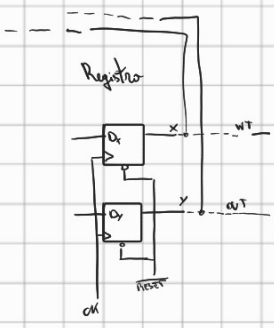


OUT → impulso "1" di durata 1 ciclo di clock alla transizione 0→1 di IN

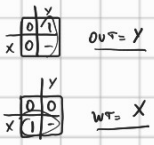
WT → si attiva a "1" se la durata in IN=1 è maggiore

di un ciclo di clock e si blocca





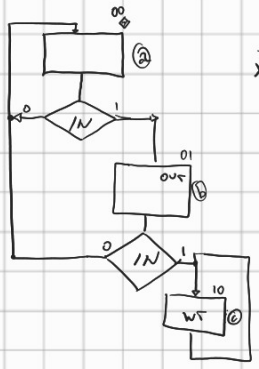
R.C. Uscita



$OUT = Y$

$WT = X$

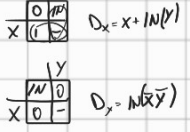
ASM





ING	Stato Attuale		Stato Successivo		
IN	X	Y	X	Y	
a	0	0	0	0	ⓐ
	1	0	0	1	ⓑ
b	0	0	1	0	ⓑ
	1	0	1	0	ⓒ
c	0	1	1	0	ⓒ
	1	1	0	0	ⓐ
?	0	1	1	–	–
	1	1	1	–	–

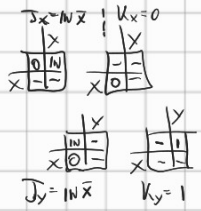
R.C. Stato Successivo



$D_X = X + IN(\bar{Y})$

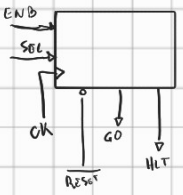
$D_Y = IN(\bar{X}\bar{Y})$

Transizione	J	K
0 → 0	0	–
0 → 1	1	–
1 → 0	–	1
1 → 1	–	0



$J_X = IN\bar{X}$; $K_X = 0$

$J_Y = IN\bar{X}$; $K_Y = 1$

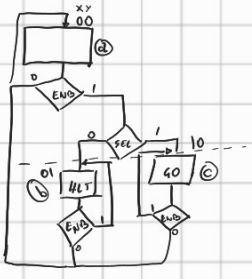


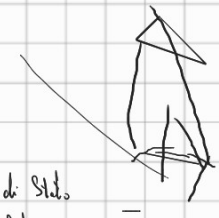
La MSF attende che

ENB Assume il valore di "1" SEL decide se MSF deve attivare GO (SEL=1) oppure HLT (SEL=0)

GO, HLT rimangono attivi per tutto il tempo in cui rimane attivo ENB

ENB torna a "0" MSF ritorna in attesa

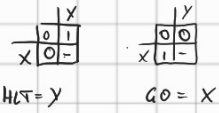




Registro di Stato a 2 bit

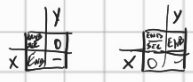


Reti Combinatorie delle uscite



$HLT = Y$ $GO = X$

Reti Combinatorie dello Stato Successivo



$D_X =$ $D_Y =$